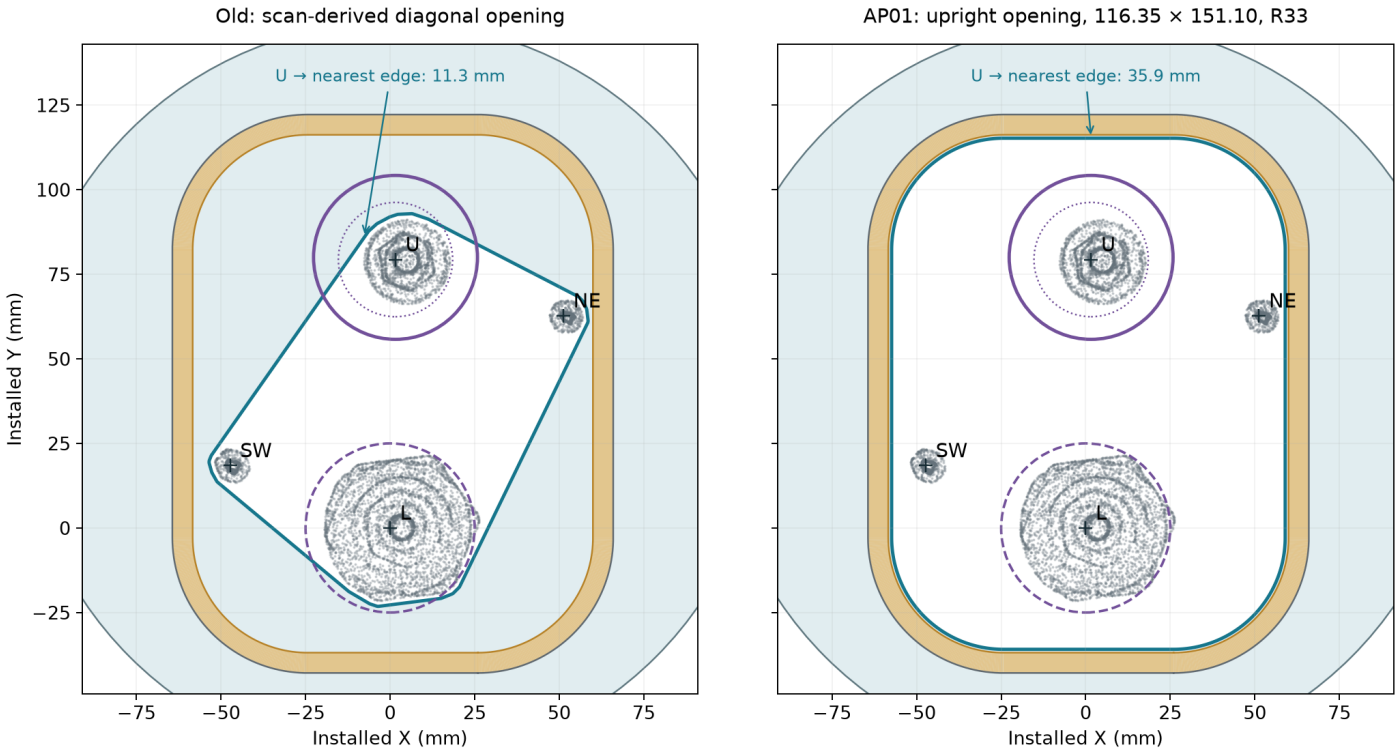
Aperture clearance around the controls

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Give the connections room, while keeping the foam on metal



Gold: approximate 6 mm edge-flush foam • Purple solid: measured Ø48.46 upper inner opening
Purple dotted: Ø33.79 cover opening • Dashed lower Ø50 is unmeasured • Gray: captured forward hardware

Projected footprints at assumed placement; different depths. A projected overlap alone is not a physical collision.

Recommended next dry-fit candidate: centered upright opening 116.35 x 151.10 mm, R33. Circular outer sizes remain C1 diameter 220 mm and C2 diameter 214.15 mm; thickness trials 1.524 or 3.175 mm. No locating tabs or separate stud holes.

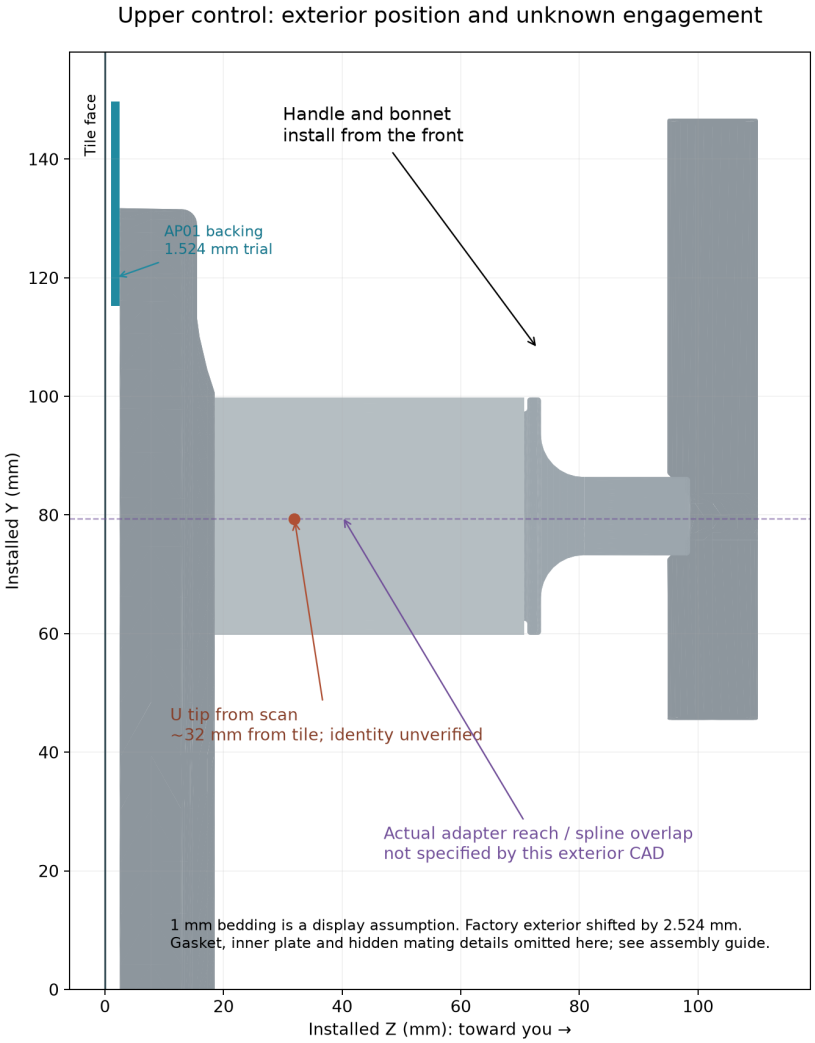
Recomputed: 51,929 clipped forward-surface triangles; no aperture conflict for either shape. Upright contains the old aperture with 0.446 mm minimum extra gap. This is a captured-surface result, not an accuracy bound.

Foam correction: outer foam edge meets metal, as confirmed. A uniform 6 mm band with the assumed R40 plate corners leaves a nominal 1 mm extra strip before the new opening. Actual corner geometry, minimum band width, alignment and compression remain to be checked.

The upper 48.46 mm circle is your measured inner opening, not a measured hub or adapter diameter. The lower 50 mm circle remains a display assumption. Original scans and six 3D tip-to-tip readings are unchanged.

The remaining check is axial engagement

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Factory sequence: attach the appropriate adapters to the stems, seat the inner plate over the studs, fit the cover, and secure it with handle assemblies on the inner-plate hubs. The complete knobs install from the front. A clear aperture does not determine spline or screw engagement.

Thickness check: added outward shift is backing thickness plus installed bedding, adjusted for any change in gasket compression. A 1.524 mm sheet with 1 mm assumed bedding shifts the factory plane 2.524 mm. The 3.175 mm sheet with the same bedding shifts it 4.175 mm. Neither is a manufacturer-approved tolerance.

Dry fit with the real components: verify backing insertion without contact, continuous gasket landing, inner plate and cover seating, both adapters properly seated and secured, intended handle-hub fastening, full operating travel without rubbing, and removal/drainage. Record the tested thickness and bedding; do not infer engagement from the U/L points.

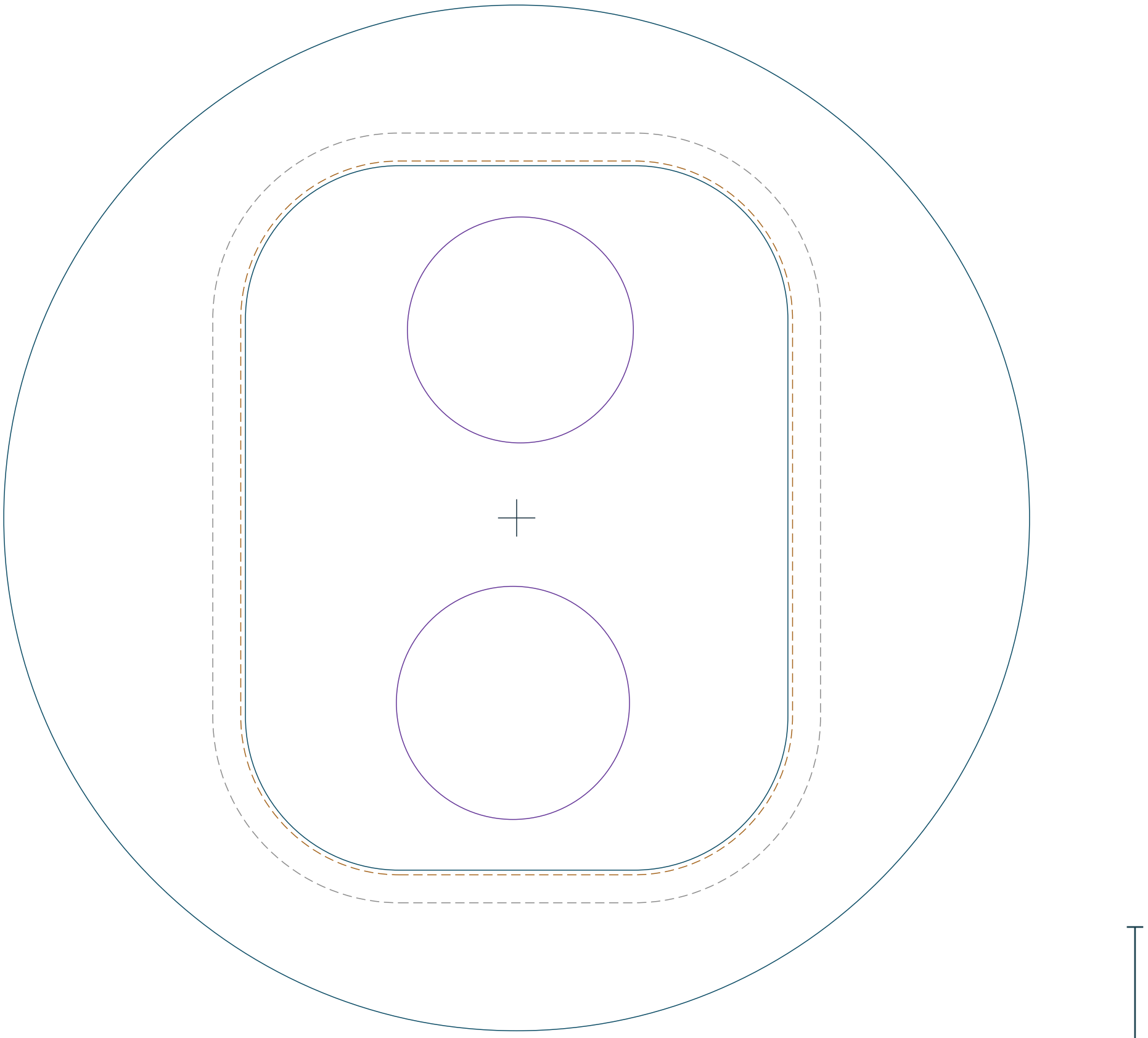
Wood relief is the user-authorized preferred approach if the trim rods need to retract and other valve/stem depths are acceptable. It does not change stem position. Required pocket dimensions and retained stud engagement remain unspecified. Existing upper-stem rough-in comparison remains conditional on bare-tip identity and scale.

Sources: Kohler 1041504-2-F pp. 7-8 (resources.kohler.com/onlinecatalog/pdf/1041504_2.pdf); official T14489-4 OBJ exterior; user R005 readings and October 15 foam-edge/contact clarification. Full inputs, limits and hashes: analysis.json beside this PDF.

Full-size template: outer diameter 220 mm

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PRINT AT 100% / ACTUAL SIZE. Disable fit-to-page. Measure both 100 mm bars before using. This paper template checks XY only; use a thickness-correct mockup for the depth/engagement test.

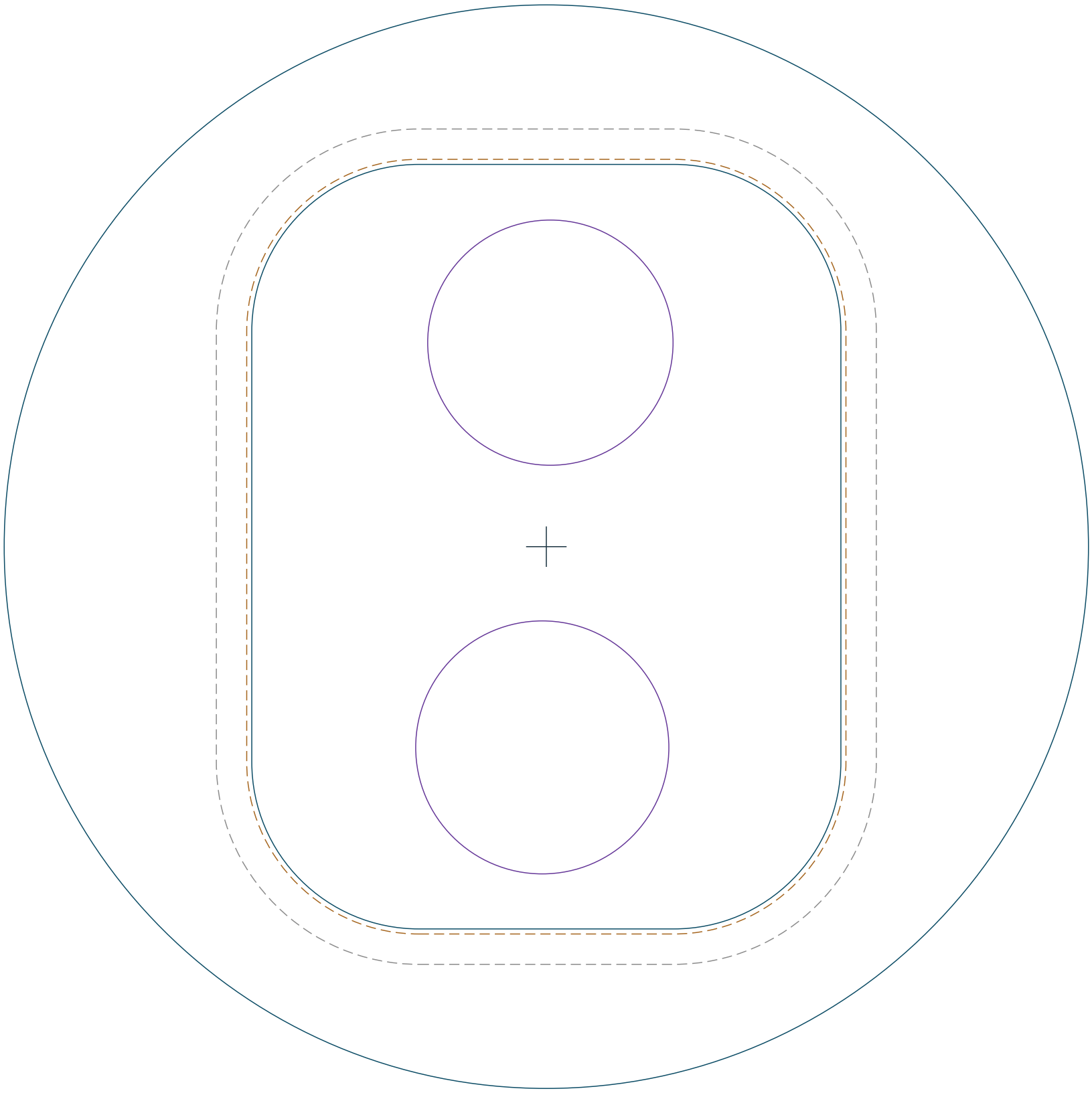


Blue: cut profile. Gray dashed: assumed inner metal outline. Gold dashed: nominal inner edge of 6 mm foam. Purple: inner-opening footprints (lower diameter unmeasured). Center cross: inferred factory plate center. All outlines upright; physical registration unverified.

Full-size template: outer diameter 214.15 mm

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PRINT AT 100% / ACTUAL SIZE. Disable fit-to-page. Measure both 100 mm bars before using. This paper template checks XY only; use a thickness-correct mockup for the depth/engagement test.



Blue: cut profile. Gray dashed: assumed inner metal outline. Gold dashed: nominal inner edge of 6 mm foam. Purple: inner-opening footprints (lower diameter unmeasured). Center cross: inferred factory plate center. All outlines upright; physical registration unverified.

